

Electrical Submersible Pump (ESP)

COMPONENTS (EQUIPMENT DESCRIPTION)

PTP- 6.1

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Electrical Submersible Pump (ESP)

COMPONENTS (EQUIPMENT DESCRIPTION)

1. Introduction

The driving force, which displaces oil from a reservoir, comes from the natural energy of the compressed fluids in the reservoir. The energy that makes the well to flow results from a reduction in pressure between the reservoir and the wellbottom. If the pressure reduction between the reservoir and the surface producing facilities is large enough, the well will flow naturally to the surface using the natural energy supplied by the reservoir.

When the natural energy associated with oil will not produce a pressure differential between reservoir and wellbore sufficient to lift reservoir fluids to the surface and into surface facilities, the reservoir energy must be supplemented by some form of artificial lift.

The ESP is considered an effective and economical means of lifting large volumes of fluids from great depths under a variety of well conditions. The submersible pump companies, along with the major oil companies, have gained considerable experience in producing high viscosity fluids, gassy wells, high temperature wells, etc. With the improved technology, wells that were once considered non-feasible for submersible pumps are now being pumped economically.

The pumping system is made up of several components. These components are: three-phase electric motor, seal section, rotary gas separator, multi-stage centrifugal pump, electric power cable, motor controller and transformers. Additional miscellaneous components will normally include wellhead, cable bands, check and drain valves. Optional equipment may include a downhole pressure and temperature sensor to monitor wellbore conditions. The ESP is suspended from discharge tubing and submerged in well fluids. A typical Electric Submersible Pump installation is shown in figure 1

The picture shows a common type of installation and is more-or-less a standard one. In this type of installation, the unit is placed above the perforation zone. The produced fluid is forced to move upward from the perforations past the motor. In doing so the produced fluid, while flowing past the motor, absorbs the heat generated in the motor and thereby cools it.



TYPICAL INSTALLATION COMPONANTS

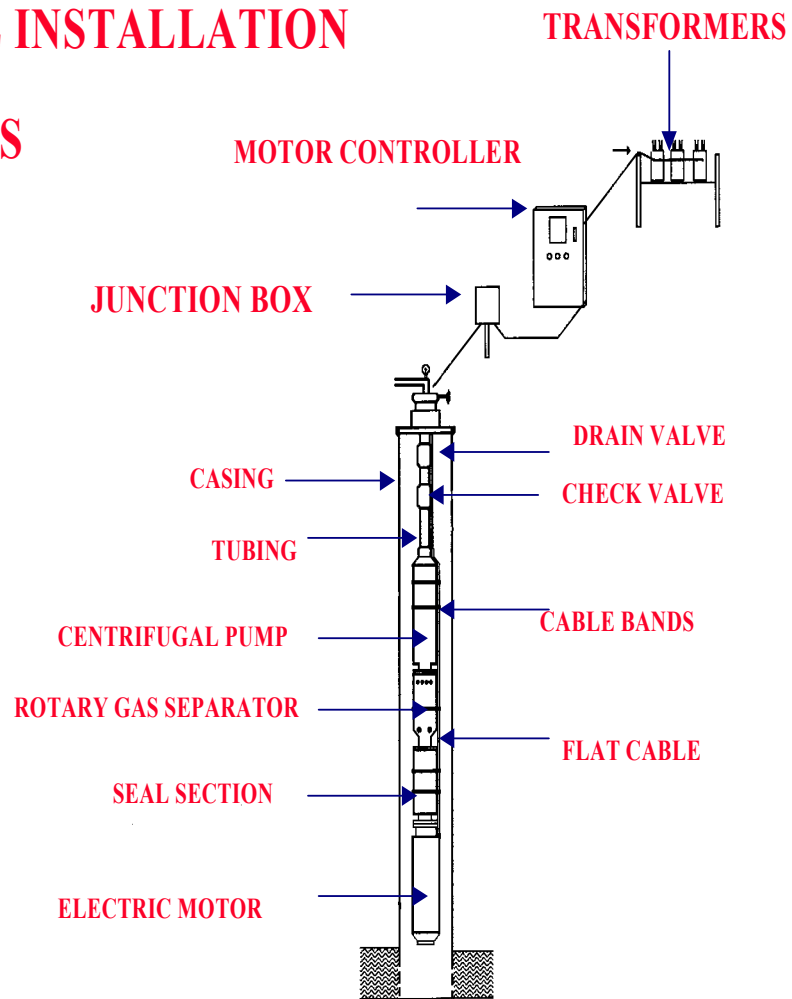


Figure 1- a (Components)

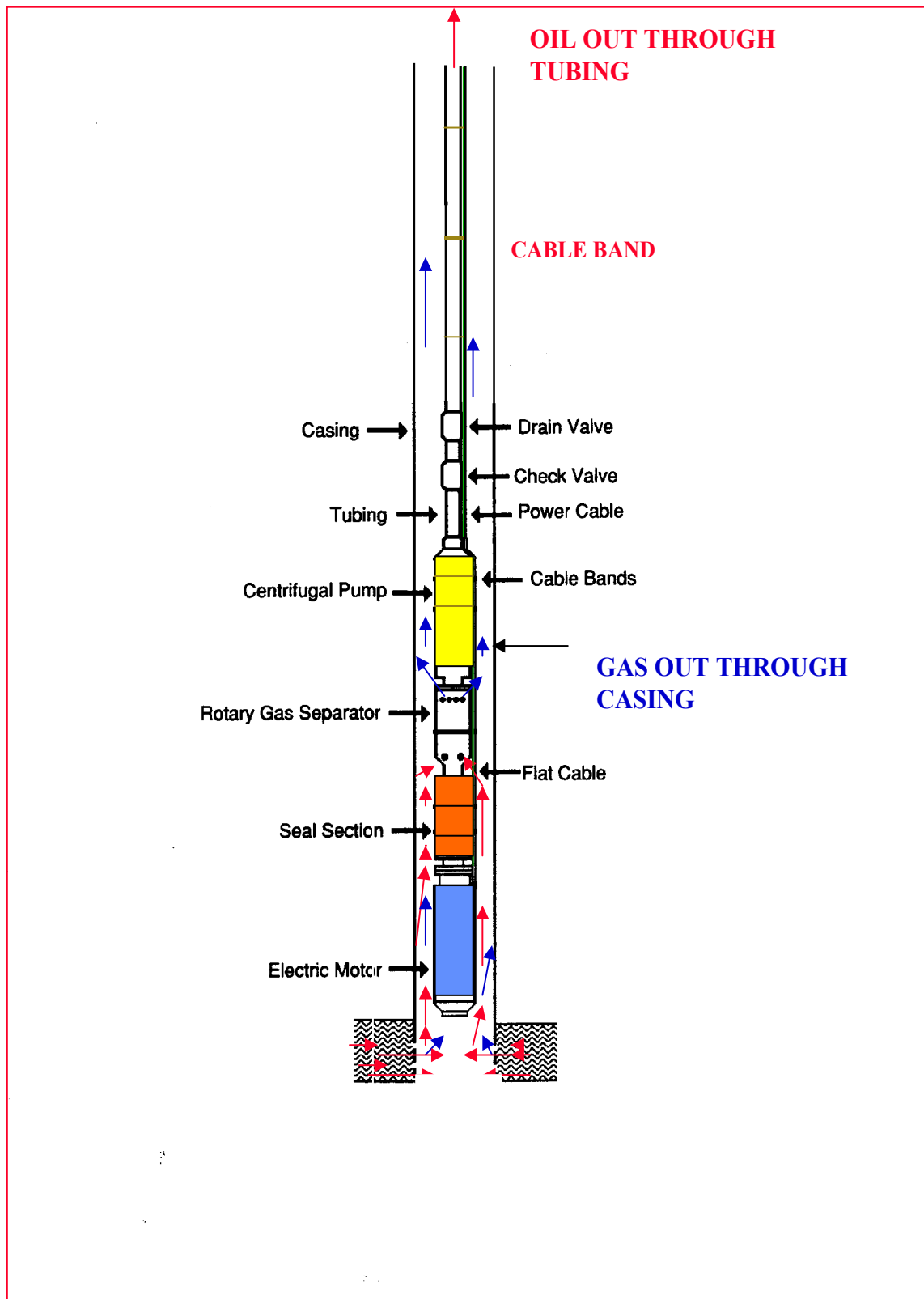


Figure 1-b (ESP Installation)

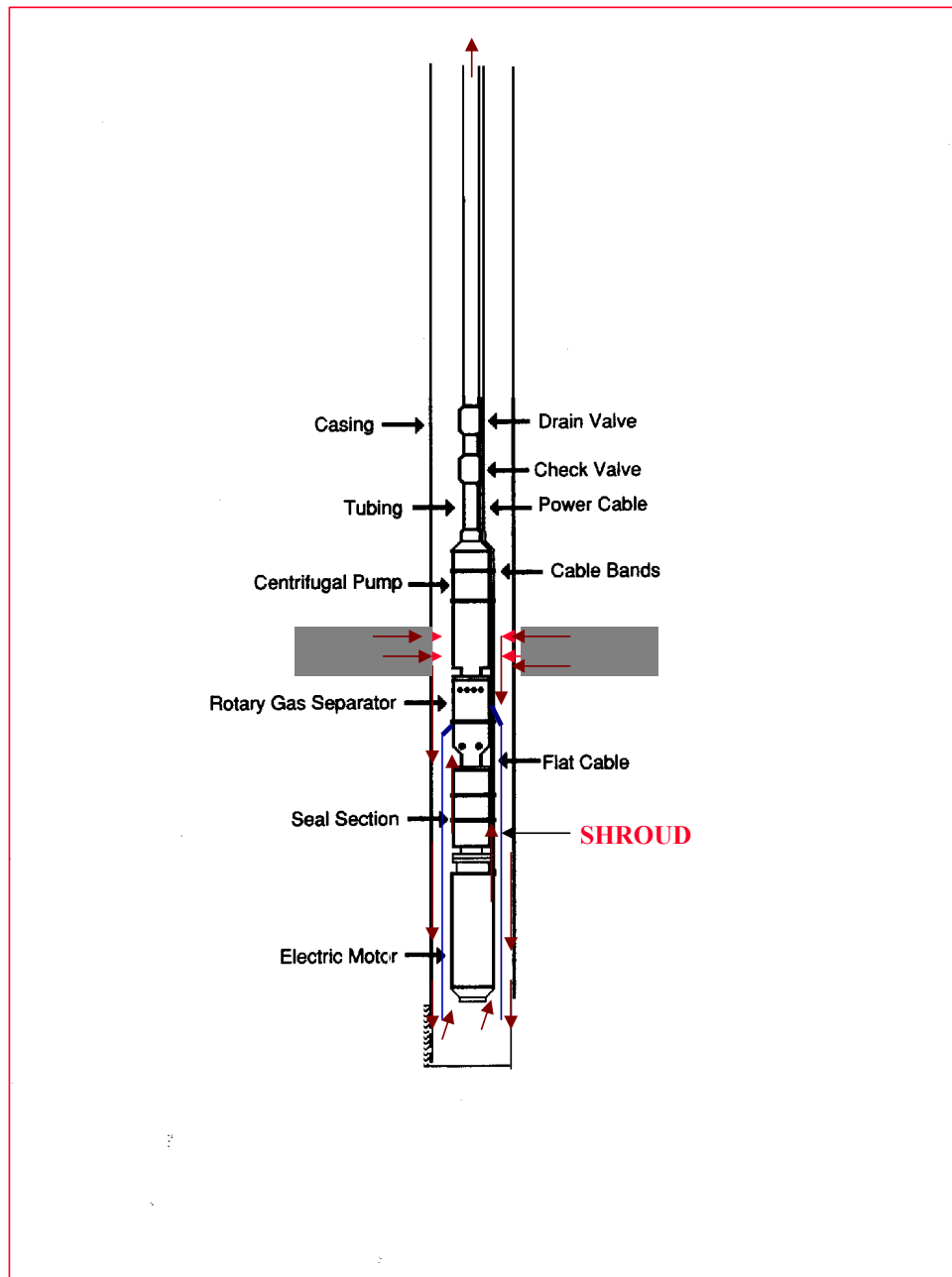


Figure 1-c (Unit is placed below the perforation)

2. Centrifugal Pump

Submersible pumps are mostly multi-staged centrifugal pumps. Each stage of a submersible pump consists of a rotating part called impeller and stationary part called diffuser. There are two components to the motion imparted to the liquid by the impeller. One motion is in a radial direction outward from the center of the impeller. This motion is caused by centrifugal force. The other motion moves in direction tangential to the outside diameter of the impeller. As a result of these two components is the actual direction of flow. The diffuser's function is to change the velocity energy into pressure energy.

There are two general types of design for submersible pumps. The smaller flow pumps are mostly of radial flow design. Figure 2-a shows such a stage configuration. In this type the impeller discharges the fluid in mostly a radial direction. As the pumps reach design flows of approximately 300 M3/d in the 400 series pumps and 550 M3 /d in the larger diameter pumps, the design changes to a mixed flow. Figure 2-b shows this configuration. The impeller in this stage design imparts a direction to the fluid that contains substantial axial direction as well as radial direction.



ELECTRICAL SUBMERSIBLE PUMP (ESP)

CENTRIFUGAL PUMP STAGES

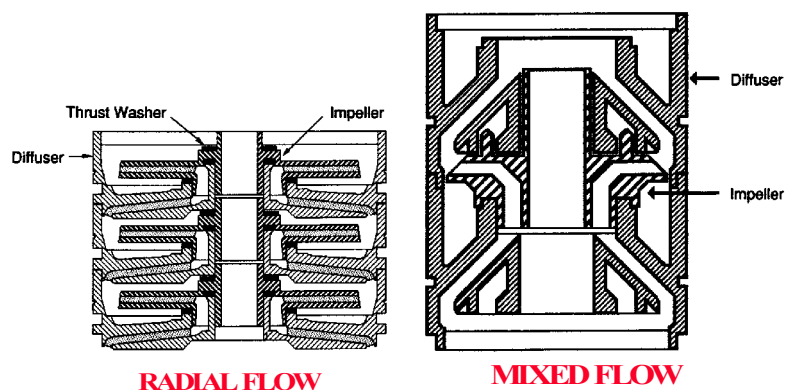


Figure 2-a Figure 2-b

In many of the pump designs, the impeller is free to float axially on the shaft and the individual impeller stage is absorbed on specially designed pads found on the diffuser. A specially designed thrust bearing contained in the seal section carries only the thrust of the pump shaft. This configuration is called a floating stage design. The benefit of this design is that many stages can be stacked together without having to fix the impellers axially on the shaft with precise alignment. As a result, pumps can be manufactured having hundreds of individual stages.

As pump diameter increases beyond about 6 inches (150 mm), the impellers are locked to the pump shaft in the axial direction and no thrust is absorbed by the impeller rubbing on the diffuser. As a result, the thrust bearing contained in the seal section must carry the impeller thrust in addition to shaft thrust. This configuration is referred to as a fixed impeller or compression pump design. The impellers are of a fully enclosed curved vane design, whose maximum efficiency is a function of the impeller design and type and whose operating efficiency is a function of the percent of design capacity at which the pump is operated. The mathematical relationship between head, capacity, efficiency and brake horsepower is expressed as:

$$\text{BHP} = Q \times H \times \text{Specific Gravity} / \text{Pump Efficiency}$$

$$Q = \text{Volume}$$

$$H = \text{Head}$$

The diameter and configuration of the pump impeller determines the amount of acceleration energy that is transmitted to the fluid. The pump size will be determined by the well casing size. The internal diameter of the impeller is dependent on the outside diameter of the shaft, which must be strong enough to transmit power to all stages of the pump. Submersible centrifugal pumps are manufactured for various sizes of well casings.

Stages are designed to have a downward thrust force on the impeller through the operating range. This force would vary from low at maximum operating with increasing force toward minimum operating. It is designed in this manner so as to provide stable hydraulic operation. Therefore, the pump should be operated within the stages recommended operating range to provide optimum life. Pumps operated outside

this range will have a reduced life and may have a detrimental effect on other ESP components.

The discharge rate of a submersible centrifugal pump depends on the speed (rpm), stage design, the dynamic head against which the pump is operating, and the physical properties of the fluid being pumped. The total dynamic head of the pump is the product of the number of stages and the head generated by each stage. Figure-3 is a typical 60 hertz, single stage, centrifugal pump performance curve showing the recommended operating range, along with other pump characteristics.

ELECTRICAL SUBMERSIBLE PUMP (ESP)
PUMP CHARACTERISTIC CURVE

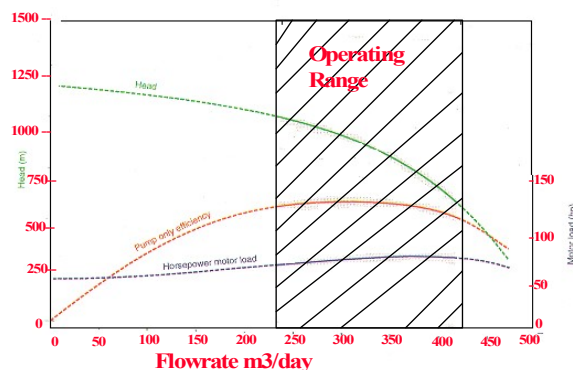


Figure -3

3. Rotary Gas Separator

ESP's can be used in wells having a high gas-oil ratio. The capability of a centrifugal pump in handling gas without gas-locking is limited. With the attachment of a rotary gas separator, ESP's can work in high gas-oil ratio wells. The separators use centrifugal force to separate the free gas (gas not in solution) from the well fluid before entry into the pump.

Figure- 4, represents the common design of a typical rotary gas separator. The fluid enters the separator and is forced into a rotating centrifuge chamber by the action of an inducer. Once in the centrifuge, the fluid with the higher specific gravity is forced to the outer wall of the rotating chamber by centrifugal force, leaving the gas in near

the center. The gas is separated from the fluid by the splitter and is exhausted back to the annulus of the well where it travels up the annulus. The heavier fluid is directed into the intake of the pump where it is pumped to the surface.

Typical gas separator efficiencies range from 80% to 95%. The system efficiency is affected by fluid volumes, composition and properties. Gas separator assemblies are often connected in tandem to improve the overall efficiency in high gas applications.

ELECTRICAL SUBMERSIBLE PUMP (ESP)

ROTARY GAS SEPARATOR

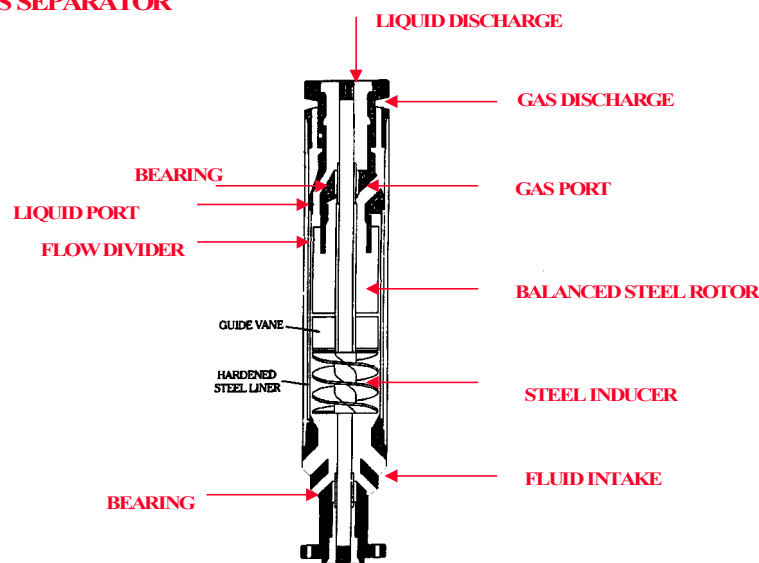


Figure - 7

4. Seal Section

The seal section connects the drive shaft of the motor to the pump or gas separator shaft and performs the following important functions:

- (1). Seal section is to allow for expansion of the dielectric oil contained in the rotor gap of the motor. Temperature gradients resulting from both the ambient and motor temperature rise will cause the dielectric oil to expand. This expansion is taken in the seal.
- (2). Seal section is to equalize the casing annulus pressure with the internal dielectric motor fluid. This equalization of pressure across the motor helps keep well fluid from leaking past the sealed joints of the motor. Well fluids which get into the motor will cause early dielectric failure. The rubber bladder shown in figure -5, as well as the labyrinth style chambers, allows this pressure equalization to take place.

(3). Seal section is to isolate the well fluid from the clean dielectric motor fluid. As previously stated contamination of the motor insulation with well fluid will lead to insulation failure. The seal section contains multiple mechanical shaft seals which keep the well fluid from leaking down the shaft. The rubber bladder provides a positive barrier to the well fluid. Labyrinth chamber provide fluid separation based on the difference in densities between well fluid and motor oil. Any well fluid that gets past the upper shaft seals or the top chamber is contained in the lower labyrinth chambers as a secondary protection means.

(4). Seal section is to absorb the downthrust of the pump. This is achieved by a sliding thrust bearing. The bearing utilizes a hydrodynamic film of oil to provide lubrication to the bearing during operation. The downthrust is the result of the pressure developed by the pump acting across the cross sectional area of the pump shaft and residual thrust transferred from each individual pump stage impeller to shaft. Figure -5 shows a cross-section of the seal section.



ELECTRICAL SUBMERSIBLE PUMP (ESP)

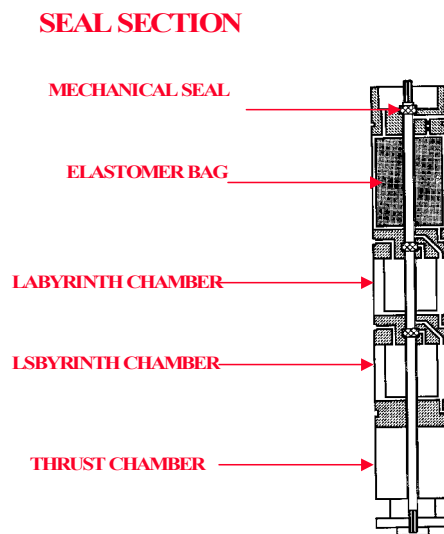


Figure - 5

Seal Section Thrust Bearing

Thrust bearing designs are of the tilting pad type. The bearing consists of 6 to 9 individual shoes which are mounted on a pedestal located at the center of the shoe. The pedestal is not so rigid that it cannot deflect and in effect this is what happens.

The operation of any bearing, is dependent up on the rotating member building and maintaining an oil film across the bearing surface. In the case of a tilting pad bearing, the thrust runner pulls fluid with it across the pad. In either case, an oil film is built and maintained otherwise failure of bearing occurs. The thrust bearing pad's deflect at the entrance edge, thus allowing the oil film to be built. An oil film can be maintained for only a limited distance due to effects of viscosity, load, temperature, etc.

5. Electrical Submersible Motor

The ESP motors are two pole, three-phase, squirrel cage, and induction type. These motors are filled with a highly refined mineral oil with high dielectric strength. The design and operating voltage of these motors can be as low as 230 volt or as high as 5,000 volt. Amperage requirement may vary from 12 to 200 amps. The required horsepower is achieved by simply increasing the length of the motor section.

The motor is made up of rotors, usually about 12 to 18 inches in length, that are mounted on a shaft and located in the electrical field (stator) mounted within the steel housing. The larger single motor assemblies will approach 33 feet in length and will be rated up to 400 horsepower while tandem motors will approach 90 feet in length and will have a rating up to 750 horsepower. Figure -6 displays the stator, rotor and shaft assembly of a submersible motor.

ESP MOTOR

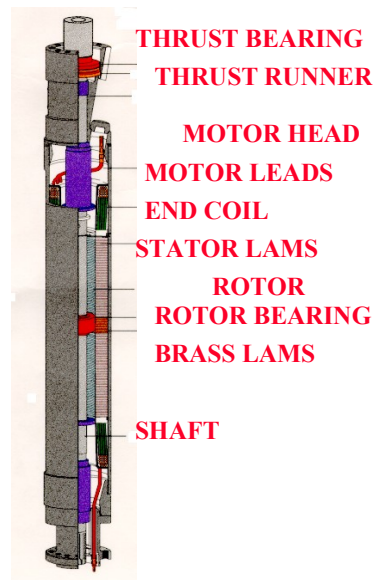


Figure- 6

The stator is made up of a group of individual electromagnets arranged such that they form a hollow cylinder with one pole of each electromagnet facing toward the center of the group. While no physical movement of the stator takes place, electrical movement is created by progressively changing the polarity of the stator poles in such a manner that their combined magnetic field rotates. In an AC motor, this is easily achieved since reversing the current each half-cycle will automatically change the polarity of each stator pole.

The rotor is also composed of a group of electromagnets arranged in a cylinder with the poles facing the stator poles. The rotor rotates simply by magnetic attraction and repulsion as its poles attempt to follow the rotating electrical field being generated by the stator. There is no external connection to the rotor, the current flow through the rotor's electrical poles is induced by the magnetic field created in the stator. The speed at which the stator field rotates is the synchronous speed. This speed can be computed with this equation:

$$N = 120 \times f / p$$

Where:

N = speed in revolutions per minute.

f = frequency in cycles per second.

P = number of magnetic poles within the motor.

Since the number of poles the stator contains is determined by the manufacturer, and are not easily changed, the only feasible way of changing the speed of the stator magnetic field is by changing the applied frequency. The rotor will attempt to follow the stator magnetic field, although due to the resistive component in the rotor conductors it lags behind. This lag is usually specified by the manufacturer and is known as slip. Even though the amount of slip varies from manufacturer to manufacturer, it is approximately 3% of the motor's synchronous speed.

6. Motor Controllers

There are two basic types of motor controllers used on ESP operations. They include the Fixed Speed Controllers, and Variable Speed Controller (VSC). Normally, all utilize solid state circuitry to provide protection, as well as a means of control, for the ESP system. The controllers vary in physical size, design, and power ratings.

Some motor controllers are simple in design, while others can be extremely sophisticated and complex while offering numerous options that were designed to enhance the methods of control, protection, and monitoring of the ESP operation. The selection of one type versus the other depends on the application, economics and the preferred method of control.

Switchboards (Fixed Speed)

The switchboard (across-the-line starter) consist of a motor starter, solid state overload and underload protection, a manual disconnect switch or circuit break circuitry and a recording ammeter. Many control systems have surface equipment for use with bottom-hole pressure and temperature monitoring equipment installed within the motor controller cabinet. Fuses are provided for short circuit protection.

External control devices should be interfaced with the controller as recommended and/or approved by the pump manufacturer to give dependable and trouble-free operation. All external control devices are connected to a time-delay which activates or deactivates the controller after a short time delay. Usual external control devices are separator high/low level controls or flowline pressure switches.

Most solid state controllers offer time delayed underload protection on all three phases, time delayed overload protection, and automatic protection against voltage or current unbalance. Underload, or some type of pump off protection, is necessary since low flow past the motor will not give adequate cooling. Circuits designed for automatic restart after shut down are normally included.

7. Variable Speed Controller (VSC)

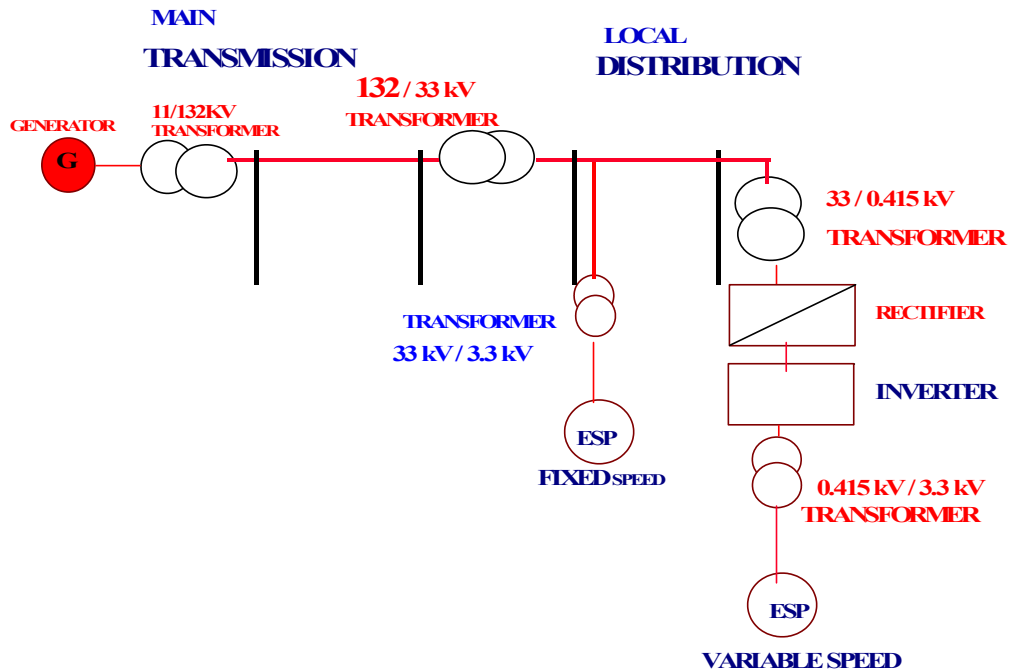
The Electrical Submersible Pump is very often not flexible when run at a fixed speed; the unit is limited to a fixed range of production rates and a fixed head output at each rate. The Variable Speed Controller (VSC) is designed to overcome these restrictions. By allowing the pump speed to be varied, the rate, the head or both can be adjusted, depending on the application, with no modification of the downhole unit.

The basic operation of the VSC is to convert the incoming 3 phase AC power, typically at 415 volts, to a single DC power supply. Then using power semiconductors as solid state switches, it sequentially inverts this DC supply to regenerate three AC output phases of pseudo-sinewave power, whose frequency and voltage are controllable. Pumping flexibility was the main purpose of applying the VSC to the ESP, but several other benefits have become apparent. Of particular interest are those that can extend downhole equipment life; soft start, automatically controlled speed, line-transient suppression and the elimination of surface chokes.

The VSC isolates the load from incoming switching and lightning transients; balances output volts to reduce motor heating; ignores frequency instability from generator supplies; compensates for brown outs or takes the unit off line; and minimizes starting stresses. In addition depending on the application, VSC's can improve overall system efficiency, reduce required size of generator, obviate the need for a choke, reduce downhole unit size and provide intelligent control functions to maximize production. All these benefits cannot be simultaneously available; however, the user must pick and chose the combination relevant to his application.



ESP POWER SUPPLY



8. Downhole Pressure and Temperature Monitor

Reservoir and pump performance data can be acquired with the use of downhole pressure and temperature monitoring systems. By correlating reservoir pressure with the pumping rate, an operator can determine when to change pump size, change injection rate or consider well workover.

Many types of downhole pressure and temperature sensors are available from submersible pump suppliers. These systems vary in design, cost, accuracy, reliability, operation and capabilities. A typical system has the capability of (1) continuously monitoring bottom hole pressure and temperature, (2) providing detection of electrical failures, and (3) can be interfaced with the Variable Speed Controllers to regulate speed.

9. Transformer

The distribution of electrical power to the oil fields is usually accomplished at an intermediate voltage, such as 33KV. Since ESP equipment operates at voltages between 1000 and 3300 volts, transformation of the distribution voltage is required.

Transformers are commonly provided in either a bank of three single phases or in a single three-phase configuration. These transformers are usually oil-filled, self cooled

units and are unusual from the standpoint that they contain a substantial number of secondary voltage taps which allow a wide range of output voltages. This wide range of voltages is required in order to be able to adjust the surface voltage for a variety of cable voltage drops that can occur due to varying setting depths.

10. Junction Box

The junction box (sometimes called a "Vent Box") performs three functions: (1) provides a point to connect the power cable from the controller to power cable from the well (2) provides a vent to the atmosphere for any gas that might migrate up the submersible power cable and (3) provides for easily accessible test points for electrically checking downhole equipment.

11. Wellhead

The wellhead is designed to support the weight of the subsurface equipment and is used to maintain surface annular control of the well. They must be equipped with a tubing head bonnet or pack-off which provides for a positive seal around the cable and the tubing, or a feed through mandrel. There are several methods available from wellhead manufacturers for providing this pack-off. Depending on the method used, the pack-off can sustain annular pressures up to 5,000 psi.

WELLHEAD

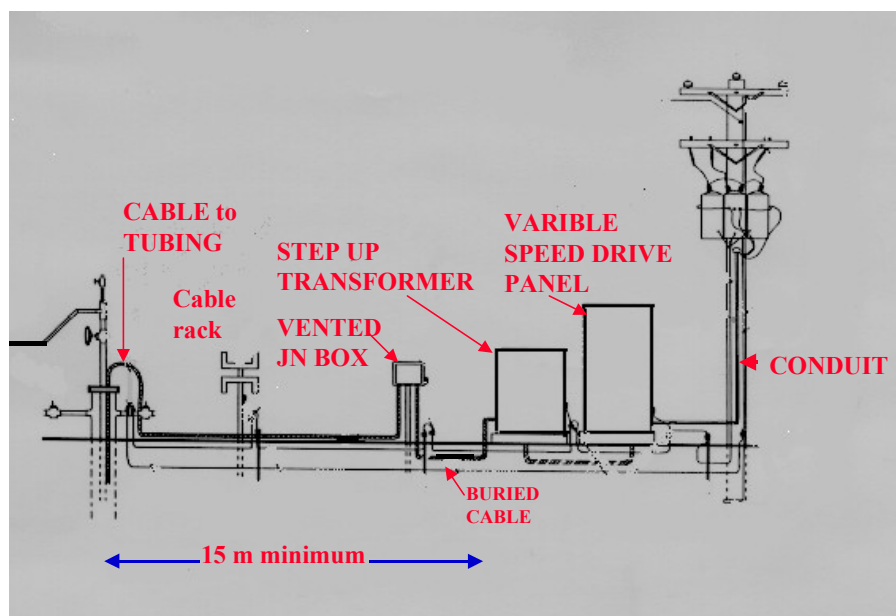


Figure – 7

Figure – 7 shows wellhead equipment of a Variable Speed Controlled ESP well. The cable should be properly laid in a cable rack or buried for safety reason. The control panel should be minimum of 15 m away from the x-mass tree.

12. Check Valve

Check valve which is usually fixed 2 to 3 joints above the pump assembly, can be used to maintain a full column of fluid above the pump. If the check valve fails or if it is not installed, leakage of fluid down the tubing through the pump can cause a reverse rotation of the subsurface unit when the motor is shut off. Start-up during the reverse rotation can result in a motor burn, cable burn or broken shaft.

If a check valve is not provided, sufficient time must be allowed for the tubing to drain back through the pump before the motor is restarted. A minimum of thirty (30) minutes is typically recommended.

13. Drain Valve

When a check valve is used in the tubing, it is necessary to install a drain valve immediately above the check valve to prevent pulling a wet tubing string. If a check valve is not provided, there is no reason to provide a drain valve, as the fluid in the tubing will generally drain through the pump while pulling.

14. Backspin Relay

In some cases installation of a check valve is impractical. For example, if you were anticipating that the subsurface equipment could plug up with scale, sand, asphaltines, etc., it may be desirable to pump produced fluid, acids or other chemicals down the tubing to remove the foreign material. With a permanent check valve installed in the tubing, this would not be possible.

There are electronic devices available that can detect a pump backspinning and will prevent restarting an ESP during this condition. The backspin relay electronics are provided in the controller and the unit is connected electrically to the ESP power cable. When the ESP system shuts down, and the produced fluid is allowed to fall back down the tubing, power is being generated by the motor and this power can be detected by the backspin relay, preventing any attempt to restart until this condition stops.

15. Centralizer

Centralizers are used in ESPs to place the equipment in the centre of the wellbore, especially useful in deviated wells, to eliminate external damage and insure proper cooling of the equipment. Many types of centralizers available in the industry designed to protect the ESP cable and prevent cable damage due to rubbing.

In corrosive conditions where coatings are used to protect the O.D. of the ESP equipment, centralizers can be very effective in preventing mechanical damage to the coating during the installation of the equipment. If the coating is damaged during

installation, the corrosion accelerates at the point of damage and the life of the equipment will be affected.

16. Cable

Electric supply is transmitted to the submersible motor by banding a specially constructed three phase electric power cable to the production tubing. This cable must be small in diameter, well protected from mechanical damage and impervious to deterioration of its physical and electrical properties by the hot corrosive well environments.

Cables are available in a wide range of sizes, which permits matching to motor requirements. They are either round or flat configurations using galvanized steel, stainless steel, or monel armor capable of withstanding the hostile environments of an oil well or water well. All cables are made to strict specifications using specially formulated materials for different operating environments

CABLE

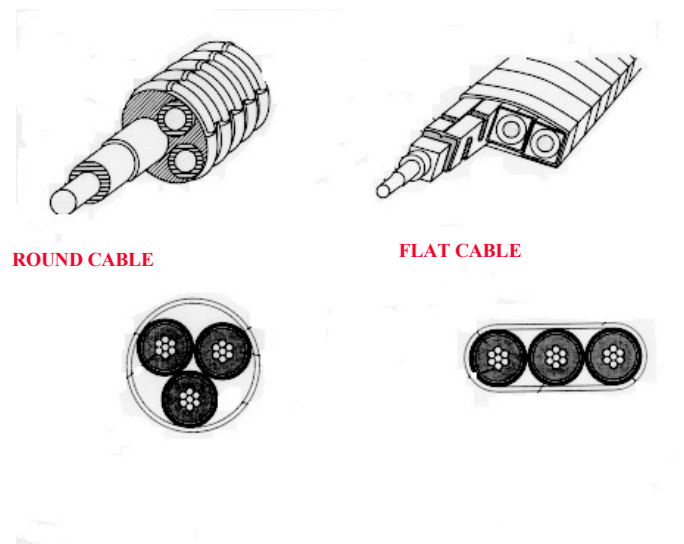


Figure - 8

17. Cable Bands

Cable bands are used to strap the power cable to the tubing. One band per 5m interval is normally used. Cable bands are also used to strap the motor flat cable to the pump and seal. Use of one band per 50cm, protected with flat cable guards, is recommended for maximum support and protection. Cable bands are made of carbon steel, stainless

steel and monel. The materials also vary in width and thickness, providing additional strength and support.

CABLE BAND



DIFFERENT TYPES OF BANDS

Figure- 9